

Education

New York University, New York, NY

Aug 2024 – May 2026

Master of Science in Computer Engineering (GPA: 3.9/4.0)

Coursework: Machine Learning, Deep Learning, Big Data, System Design, Data Structures, Advanced Java

Experience

- **Amazon** May 2025 – Aug 2025
Software Development Engineer Intern Seattle, US
 - Built **LLM-based retrieval pipelines (RAG)** with agentic workflows and a **Model Context Protocol (MCP) server** for internal tooling; improved contextual accuracy for production inference use cases.
 - Engineered backend services with **OOP design patterns**; owned features from design through deployment, monitoring, and defect resolution with **Mockito** test coverage and **CI/CD**.
 - Provisioned **serverless ML-adjacent infrastructure** using **AWS CDK** (API Gateway, Lambda, DynamoDB, S3, IAM) with multi-stage deployments across regions.
- **New York University (Novel AI Technologies, Inc.)** Aug 2024 – Present
Software Developer New York, US
 - Deployed cloud-hosted **AI/ML models** for health metric analysis and image processing on an NSF-funded mobile platform processing real-time wearable data.
 - Engineered **Python ETL pipelines** transforming **NoSQL** into **PostgreSQL** with concurrency controls; processed **250K+** records and powered **AWS QuickSight** analytics dashboards.
 - Developed an insurance analytics platform (**React, Python, Node.js, REST APIs**) on **Docker/EC2**; implemented RBAC and row-level security for production users.
- **CampusMesh** Jan 2026 – Present
Software Engineer New York, US
 - Worked on **embedding-based semantic search** and **RAG pipelines** for CampusIQ (AI Assistant), improving retrieval accuracy for structured and unstructured campus data.
 - Developed **Python/Node.js microservices** on **AWS** (Lambda, API Gateway, S3) with **MySQL** and **DynamoDB** backing stores across local through production environments.
- **PricewaterhouseCoopers (PwC)** Aug 2021 – Aug 2024
Senior Software Engineer Bengaluru, India
 - Deployed an **NLP-based OCR neural model** on **Azure** to automate structured data extraction, reducing processing time by **75%**.
 - Diagnosed complex production incidents across distributed **Azure** services; led code reviews and mentored engineers on **software design principles**.

Projects

- **RoBERTa Fine-Tuning with Custom LoRA** | *Python, PyTorch, Hugging Face, PEFT, AGNews*
 - Architected a **parameter-efficient fine-tuning (PEFT)** pipeline with custom **LoRA** for **RoBERTa-base** news classification; ran hyperparameter sweeps over rank and learning-rate schedules to maximize validation accuracy under memory constraints.
 - Applied **AdamW** optimizers and linear decay schedulers; achieved **83.45%** accuracy with under **1M trainable parameters** via rank-decomposition matrix injection.
- **InviLite (ML Incident Analysis Pipeline)** | *Python, AWS, SageMaker, Bedrock, FAISS, SVM*
 - Built a **RAG system** (SVM + FAISS + **Bedrock LLM**) lifting root-cause classification accuracy from **56% to 92%**; architected governed ingestion pipelines (Kinesis, Lambda, S3, Glue, DynamoDB) for enterprise data.

Skills

- **Programming Languages:** Python, Java, C++, TypeScript, JavaScript (ES6), SQL
- **AI/ML:** PyTorch, LLM Fine-tuning, RAG (LangChain, FAISS), NLP, Deep Learning, Hugging Face, scikit-learn
- **Web & Backend:** Node.js, REST API, Microservices, React, GraphQL
- **Databases:** PostgreSQL, MySQL, DynamoDB, MongoDB
- **Cloud & DevOps:** AWS (CDK, Lambda, EC2, S3, SageMaker, Bedrock), Azure, Docker, Kubernetes, CI/CD, GitHub Actions
- **Tools & Platforms:** Git, Linux, Cursor, GitHub Copilot, Jupyter, Postman, Pandas, NumPy

Certifications

AWS Cloud Practitioner

Stanford Machine Learning (Coursera)

Azure Data Fundamentals